

Wanglong Lu

Senior Data Scientist | Applied Multimodal Intelligence

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Profile

Applied ML researcher and Senior Data Scientist at Nasdaq, with a Ph.D. in Computer Science. Works on financial ML, multimodal image editing, and efficient adaptation, with experience deploying ML workflows on AWS.

First-author **IEEE TNNLS** and **TVCG** publications; **NeurIPS** research integrated into **Hugging Face PEFT**.

Technical Skills

Core: Python, PyTorch, SQL, Git, Bash, C++, CUDA; experimental design and feature engineering.

ML & GenAI: Latent diffusion, Transformers, multimodal AI, PEFT/LoRA, Hugging Face, RAG, agentic tools.

ML systems: FastAPI, Mangum, PostgreSQL; AWS SageMaker, Bedrock, Lambda, S3, Docker, SLURM.

Professional Experience

Nasdaq / Nasdaq Verafin

St. John's, NL, Canada

Senior Data Scientist, AI/Analytics

Mar. 2025 – Present

- Designed cross-scale ensemble feature selection for **cheque-fraud ML (CFML)**, removing **77.2% of input features** while retaining competitive performance in evaluations.
- Deployed and integrated an **existing UI/backend prototype** for a cross-team ML training platform under development into an internal workspace; configured networking and IAM access. Platform scope includes ingestion, training, evaluation, and model governance.
- Added **PostgreSQL-backed user-state persistence** and deployed **FastAPI** services on **AWS Lambda using Mangum**, connecting application state and backend services to the workspace.
- Built **SageMaker Job Submitter**, a Python package for local scripts and dependencies across **Processing, Training, and Hyperparameter Tuning**, with **SageMaker Pipelines**. Package-based submission cut developer job-configuration time from **7 days to 1 day**, an approximately **86% reduction**.
- Developed a **Bedrock-based feature-formula validator** with structured logical-consistency checks and reliability scores to flag hard-coded errors in feature-engineering formulas.
- Built a **PagerDuty RAG incident-response assistant** as an internal hackathon project on the company's GenAI platform; used incident context to retrieve relevant troubleshooting guidance and resolution steps for on-call engineers.
- Mentored industry research on **cheque OCR** quality assessment and text extraction, and **tabular-data generation**, supporting pipeline design and generative-model architecture.

AI Algorithm Intern

May 2024 – Jan. 2025

- Designed diffusion-based restoration and inpainting for degraded cheque images in Python/PyTorch; ran comparative experiments against existing methods using **AWS, SLURM, CUDA, and shell scripts**.
- Developed and evaluated patch-level Gaussian latent modeling for image verification; presented research progress to **39 cross-functional attendees** at the end of the internship.

Beaufort Solutions

St. John's, NL, Canada

AI Intern

Jul. 2022 – Nov. 2022

- Designed a few-shot vision-language classifier for personalized photo-book curation; led data collection and evaluation with Python/PyTorch and AWS/SLURM experiments. Contributed to TEG (Journal of Electronic Imaging, 2024).

Zhongkong Hanlian

Hangzhou, China

AI Algorithm Intern

Dec. 2017 – Aug. 2018

- Built ResNet/DenseNet vehicle-logo recognition with Caffe, Python, and C++; created Java annotation tools and a training/evaluation dataset, and deployed models for **real-time inference at community security gates**.

Selected Research Publications

UltraDiffEdit – High-Resolution Generative Editing

First author; IEEE TNNLS 2026

- Developed tuning-free editing with multi-patch encoding, global-local denoising, and progressive sampling; demonstrated **8K editing on a single RTX 3090**. Code; video.
- Built **three editing benchmarks** (DIV2KEdit, Syn2KEdit, UHRSEdit) to evaluate high-resolution, multimodal editing; tested pretrained models including **SDXL and ControlNet without fine-tuning**.

FACEMUG – Multimodal Local Facial Editing

First author; IEEE TVCG 2024

- Designed multimodal feature fusion and self-supervised latent warping for **six input modalities**: text, sketches, semantic maps, color maps, exemplars, and attribute labels.
- Preserved unedited regions during local, controllable facial editing; presented the framework through the **Genesis entrepreneur training program**. Paper.

AdaMSS – Parameter-Efficient Adaptation

Co-author; NeurIPS 2025

- Contributed to multi-subspace fine-tuning; reported ViT-Large results exceeded LoRA's average accuracy across seven tasks using **15.4% of LoRA's trainable parameters**. Method **integrated into Hugging Face PEFT**. Code; PEFT integration.

DocPure – Unified Document Restoration

Co-first author; IEEE TCSVT 2026

- Co-authored prompt-free document restoration using degradation-aware structure extraction and structure-guided wavelet modulation across multiple degradations. Code.
- The unified model was evaluated on **12 datasets across four tasks**: deblurring, denoising, compression artifact reduction, and deshadowing. Benchmarks.

GRIG – Data-Efficient Image Inpainting

First author; CVM 2025

- Developed iterative residual inpainting using CNN features, Transformer reasoning, and forgery-patch adversarial training; evaluated on **10 few-shot datasets** using FID and LPIPS. Code; benchmarks.

Visual Style Prompt Learning

First author; Pattern Recognition 2025

- Developed diffusion-generated visual prompts and style-modulated aggregation for blind face restoration. Code.

Full publication record: Google Scholar and publication portfolio.

AI Engineering Prototypes & Deployment

Local Agentic Code Editor

Nasdaq internal prototype; 2026

- Prototyped a local AI code editor with an **agentic loop** that selects code writing, code review, or shell-tool execution according to the task.
- Connected **language-model APIs with local tool execution** for code generation, debugging, and iterative review in a working exploratory prototype.

Education

Ph.D., Computer Science – Memorial University of Newfoundland, Canada

2021–2025

M.Sc., Computer Software and Theory – Wenzhou University, China

2018–2021

B.Sc., Digital Media Technology – Communication University of Zhejiang, China

2014–2018

Selected Recognition & Service

Outstanding Research Award (Ph.D.), Memorial University, 2023; four granted Chinese invention patents.

Mentored **10 students** in generative vision and applied ML, including restoration and recognition projects.

Teaching assistant in Data Structures and Algorithms (2021) and Introduction to Machine Learning (2024).

Invited reviewer for **IEEE TPAMI, TIP, TMM, TCSVT**, Pattern Recognition, Knowledge-Based Systems, Neurocomputing, Applied Soft Computing, and ECCV 2026.

Presented feature-selection research to **60+ attendees** (2026); invited speaker, IEEE Newfoundland and Labrador Section AGM (2026); guest lecturer on deep generative models, Memorial University (2025).